

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

(192511387)

I Betty A. Shanika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Betty

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1) cloud Auto - scaling and Load Balancing

cloud Auto - Scaling :

Automatically increases or decreases computing resources based on traffic

Handles sudden traffic spikes during seasonal sales.

Reduce resource wastage during low traffic.

Load Balancing :

Distributes incoming user requests across multiple users.

Prevents server overload

Ensures all servers share the workload equally.

Reduces Downtime :

If one server fails, traffic is redirected to healthy servers.

Keeps the application available without interruption.

Improves User Experience

Faster page loading

Smooth shopping experience

Reduces delays and failed transaction

Role of monitoring tools

Monitor CPU, memory, network usage, and response time

Trigger auto-scaling when demand increases

Send alerts for system issues

Real-World Example:

During Amazon's Great Indian Festival or Flipkart Big Billion Days, cloud auto-scaling and load balancing handle millions of users simultaneously, ensuring uninterrupted shopping.

2) cloud storage ^{for} Backup Data

Durability:

Data is stored across multiple storage devices

protects against hardware failures

Ensures long-term data availability.

Data Redundancy:

multiple copies of data are stored in different locations.

prevents data loss during disasters.

Encryption:

Encrypts data while storing and during transmission

prevents unauthorized access.

Access Control:

uses IAM roles, passwords, and multi-factor authentication (MFA)

only authorized users can access backup files.

Cost Advantages:

uses IAM roles, passwords, and multi-factor authentication

no need to purchase expensive storage hardware

pay only for the storage used

lower maintenance costs.

Disaster Recovery Example

If a company's data center is damaged due to fire or flooding, backup data stored in the cloud can be restored quickly, ensuring business continuity.

3) content Delivery Networks (CDNs)

CDN:

Stores copies of website content on multiple edge servers.

Delivers content from the nearest server.

Reduced Latency:

user receive data from nearby locations.

Faster loading and lower network delay.

Edge Servers:

cache frequently accessed content

Reduced load on the main server

Improve response time.

Global Availability:

cloud providers maintain data centers worldwide

Users receive consistent performance from any location.

Real time Streaming

Netflix uses CDNs to stream videos smoothly.

Gaming Example:

PUBG and other online games use edge servers to reduce lag and improve gameplay.

Scalability:

CDNs can handle millions of users simultaneously by distributing traffic across multiple edge servers, ensuring smooth streaming or gaming even during peak usage.

4) cloud security and compliance

Regulatory compliance:

cloud providers support standards

like ISO 27001, GDPR, and PCI-DSS.

Help organizations meet legal requirements.

Logging:

Records all user and system activities

Helps during audits and investigation

Monitoring:

continuously monitors system activity.

Detects suspicious behaviour and security threats.

Encryption:

protects sensitive financial information

secures stored and transmitted data.

Data sovereignty:

stores data in specific geographic regions as required by law.

Ensures compliance with local regulations.

Example:

A bank encrypts customer transaction data, stores it within the required country, and uses audit logs to monitor every financial transaction securely.

5) containers vs virtual machines

container vs virtual machine:

containers share the host operating system

virtual machines require a separate operating system

containers start faster and consume fewer resources

VMs provide stronger isolation but use more memory and storage.

Role of Kubernetes:

Automates deployment of containers

managers scaling automatically

Restarts failed containers

Balances application traffic

Portability:

containers run consistently across different cloud platforms.

simplifies migration application
where between AWS, Azure, and
Google cloud.

Microservices Example:

An e-commerce application
where user login, product catalog,
payment, and order management run
as separate containerized microservices,
managed by Kubernetes for high
availability and scalability.